GAME TEST REPORT

Dangrondropa Mines

Pre-Certification Self-Assessment

PASSED WITH NOTES

Report Number	`BS-SPINNY-20260801-70EC6CF5`
Date of Report	2026-08-01
Issued By	BlueStuff Game Studio · Automated Test Engine
Report Recipient	6a43beed9eae7b49adcdc60e
Engine Under Test	`spinny`
Standard Tested Against	GLI-19 v3.0 — Interactive Gaming Systems
Classification	Pre-certification self-assessment (Non-Jurisdictional)
Software Fingerprint	`sha256:c28f83af68b4d184d4210e86e52d53577f2b4c7a…` · 30 files
Result Summary	3 passed · 0 failed · 0 warnings · 1 manual · 0 source findings

Disclaimer. This is an automated pre-certification self-assessment produced by BlueStuff Game Studio's own test engine, published unmodified — including any failures. It is tested *against* the GLI-19 v3.0 technical standard but is **not** an accredited-laboratory certification and does not imply affiliation with, or endorsement by, Gaming Laboratories International. Final certification requires an independent accredited testing laboratory.

1. Executive Summary

Every hard requirement that ran passed. Some results are provisional or need human review.

Evaluator's Assessment

The reasoned opinion of the automated evaluator, based strictly on the measured results in this report.

GLI-19 Compliance Evaluation — Spinny

Game ID: ea4b051c-ca04-410d-98d5-fc52a59a4872 | **Engine:** spinny | **Run:** 70ec6cf5 | **Date:** 2026-08-01

Overall Determination: **Inconclusive — Not Fully Tested**

The RNG subsystem is statistically sound, but the game's payout math has not been verified. Spinny **cannot be certified for real-money operation** at this time. This is not a marginal result — it is a fundamental gap: the engine produced zero payouts across 200,000 simulated spins, leaving the single most player-critical requirement (§4.7.1 RTP) unresolvable by any statistical method.

What Passed

The shared crypto-backed RNG clears all three applicable GLI-19 RNG clauses at the 99% family-wise confidence level:

Clause	Verdict	Key Statistic
§3.2.2 — Statistical quality (7 tests)	**PASS**	Fisher combined p = 0.416; 0 of 7 tests rejected (Holm–Bonferroni)
§3.2.3 — Uniform distribution	**PASS**	Float 32-bin p = 0.093; int 64-range p = 0.816
§3.2.4 — Independence	**PASS**	Fisher combined p = 0.664; 0 of 3 tests rejected

These results are based on 600,000 samples across 5 independent replicates. The RNG is fit for purpose and would not require re-testing if only the game-math layer changes.

The Blocking Issue: §4.7.1 — Return to Player (MANUAL)

The audit engine drove 200,000 spins against the registered backend and observed **no payouts**. The engine is either a stub, an integration placeholder, or is missing the payout-resolution logic entirely. The §4.7.1 verdict is MANUAL — not a borderline fail, but an untestable state. Concretely:

- **For players:** An operator cannot lawfully represent this game as having a certified RTP. Any displayed RTP figure (e.g. "96%") would be unsubstantiated.
- **For operators:** Deploying this build to real-money wallets exposes the operator to regulatory enforcement. Any jurisdiction that requires pre-certification would not accept a MANUAL verdict here.
- **For certification:** §4.7.1 is not waivable. The RTP measurement must be repeatable and deterministic before the audit can close.

Adversarial Findings

No adversarial findings were returned. This is reported as measured — it does not imply the backend was fully exercised. Runtime probes (concurrent-bet race conditions, cash-out after settlement, session fixation) require a running containerised backend; those classes were **not tested** in this run and remain open.

Recommendations (prioritised)

1. **[Blocking]** Implement and connect the payout resolver in the spinny backend engine. The `play()` path must return multiplier outcomes that the audit engine can observe. Re-run §4.7.1 with `--spins=200000` (minimum); 1,000,000+ spins is recommended for a target RTP above 95% to achieve statistical power.
2. **[Blocking before live]** Once a containerised backend exists, re-run adversarial probes to cover the runtime attack surface (double-spend, cash-out timing, session fixation). The current clean adversarial result reflects an untested scope, not a confirmed clean state.
3. **[No action needed]** RNG clauses §3.2.2, §3.2.3, §3.2.4 are clear. Do not re-test these unless the RNG implementation changes.

This report should **not** be staged for public certification publication until §4.7.1 returns a measurable verdict. The current state is suitable only as an internal development checkpoint confirming RNG readiness.

2. Scope & Evaluation Elements

The following elements were evaluated to the extent applicable to automated testing of this software:

Element	Status
Software & System Version Control	Performed
Source Code Review	Performed
Game Accounting (RTP / payout)	Performed
RNG & Statistical Analysis (GLI-19 §3.2)	Performed
Functionality Testing	Partial

3. Requirement Results

Requirement	Result	Sample	Statistic
3.2.2	PASS	600,000	`p = 9.33e-2`
3.2.3	PASS	600,000	`p = 9.33e-2`
3.2.4	PASS	120,000	`p = 4.43e-1`
4.7.1	MANUAL	200,000	–

4. Requirement Detail

3.2.2 PASS

Shared RNG source (crypto-backed, used by every engine on this template). float() scaled to 32 bins, 5 independent draws combined by Fisher's method. 7 of the 7 tests named in §3.2.2 applied. None rejected at the 99% family-wise confidence level (Holm–Bonferroni; Fisher combined p = 4.156e-1).

3.2.3 PASS

RNG output is uniformly distributed (float 32-bin: Fisher chi-square = 16.23, p = 0.0933; int 64-range: p = 0.8155).

3.2.4 PASS

Independence of the shared RNG (serial, overlaps, interplay) and of each engine's per-round outcomes (none playable). 3 of the 7 tests named in §3.2.2 applied. None rejected at the 99% family-wise confidence level (Holm–Bonferroni; Fisher combined p = 6.638e-1).

4.7.1 MANUAL

Return-to-player measured per registered engine by driving play() with the template RNG: • example: no payouts observed over 200,000 spins — stub or non-payout engine; nothing to certify. No registered engine produced payouts — this backend has no certifiable game math yet. (1 non-payout/stub engine(s) noted above.)

5. Source Code Review

No adversarial findings. The probes that were run did not surface an exploitable issue.

6. Software & System Version Control

A SHA-256 checksum was generated for each of the 30 source files in the engine under test. The aggregate fingerprint binds this report to exactly the code tested.

Aggregate fingerprint: `sha256:c28f83af68b4d184d4210e86e52d53577f2b4c7a7bea72bdd86f66a41b8dc8f7`

File	Bytes	SHA-256
`mock-rgs/server.js`	6,941	`efd3b571471ff8715098…`
`package-lock.json`	39,470	`80396a8b9dc627a8b7bb…`
`package.json`	650	`dbe06c33c23778839840…`
`src/adapters/InlineWallet.js`	3,591	`0f3d448ea065d1965b13…`
`src/adapters/RgsAdapter.js`	11,741	`bf249f586850cb72fe3d…`
`src/adapters/WalletAdapter.js`	1,827	`96abb0fa90468bbcb94a…`
`src/app.js`	2,960	`4853c5de79c0d57c1eb9…`
`src/config/index.js`	2,048	`70ca00b00159577b15f3…`
`src/engines/GameEngine.js`	3,261	`83d6fff9314cb1845d45…`
`src/engines/example/ExampleEngine.js`	2,827	`b146f8322209c3764e3c…`
`src/engines/index.js`	1,419	`e3b360a4589c591a6378…`
`src/middleware/errorHandler.js`	936	`e3c1912b7fe62a86a9cd…`
`src/models/index.js`	4,175	`763a60ca94e6cec820e5…`
`src/routes/config.js`	957	`a4c5adc3cd5cceaabe52…`
`src/routes/game.js`	1,742	`07682a3da73b014775f1…`
`src/routes/session.js`	1,309	`f3d8ea81cc657ab65445…`
`src/server.js`	1,010	`cda1ea92956411d4dd94…`
`src/services/AuditService.js`	1,205	`2887a68af4ba2500ecb7…`
`src/services/GameService.js`	13,281	`2111fc9a04afb7c07507…`
`src/services/SessionManager.js`	3,501	`9a133289e6702199ceb7…`
`src/store/index.js`	554	`90e0279b64368867bc05…`
`src/store/memoryStore.js`	1,524	`33fcc249eed7130402f9…`
`src/store/mongoStore.js`	1,177	`89d67140ee3156976ce5…`
`src/utils/errors.js`	1,710	`0d4ee388580860da1aba…`
`src/utils/ids.js`	1,036	`19816c83e246299f080e…`
`src/utils/logger.js`	1,938	`6ab4fc4d1af74e62a7a4…`
`src/utils/money.js`	996	`6b633695f0ace232e6cd…`

`src/utils/rng.js`	1,106	`6a75832219a1d9540812…`
`tests/lifecycle.test.js`	11,972	`e4c304aebdb95a95b6f1…`
`tests/rgs-adapter.test.js`	8,944	`d75b50b824c88f8ab9e7…`

7. Verification & Reproduction

The test engine has no dependencies and no LLM in its numeric path — every statistic is computed by reviewable in-tree code. This run reproduces with:

```
node src/cli.ts --engine=spinny --repo=D:\Projects\Devops\MCP\mcp-server\.repo-cache\games\bluestuff-io__smokestack-original-games\BackendEngines\spinny --samples=100000 --spins=200000 --replicates=5 --seed=gli-1785584499864
```

The seven statistical tests are the ones GLI-19 §3.2.2 names explicitly (chi-square, overlaps, coupon collector, runs, interplay correlation, serial correlation, duplicates), evaluated collectively at 99% family-wise confidence and combined across independent replicate seeds.

8. Test Environment

Test engine	SportsITDev/gli-audit-engine
Runtime	Node v24.14.0
Seed	`gli-1785584499864`
Started	2026-08-01T11:41:40.006Z
Finished	2026-08-01T11:41:44.064Z
Generated by	gli-audit-engine game-tester agent